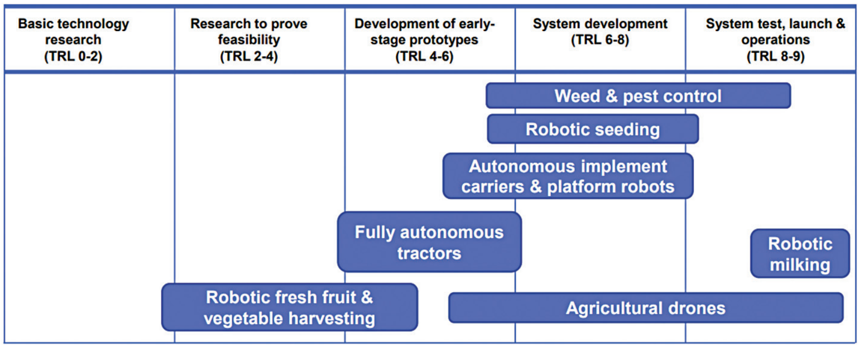

Fig. 4: Technology readiness of different types of agricultural robots (Credit: IDTechEx)

Fig. 5: Progression of tractor development. The extreme left shows a conventional heavy manned tractor while the extreme right shows a swarm of light robots that can cover more field than conventional tractors (Credit: IDTechEx)

> Data ownership and management is an ignored issue that encompasses all the other challenges. Whether the data belongs to farmers, landowners, or robot manufacturers is not clear yet. There is an ongoing debate around this. No common consensus has emerged yet. Future policies regarding the use of agricultural robotics will be largely based around the outcome.

(ADAS) and autonomous vehicles (AVs), an evolutionary pathway is ideal before self-driving tractors emerge. The first type is guidance tractors that help farmers drive tractors with less fatigue. It should also enable farmers to work under dark lighting conditions and ensure the entire field gets covered. A robust GPS technology is the best tool for assistance.

The next logical step would be the introduction of 'follow-me' tractors or a series of unmanned tractors controlled by a manned vehicle that effectively boosts driver productivity without facing any technical or legal complexity of creating a fully autonomous farm vehicle.

And, finally, there are fully autonomous driverless systems where operators control the vehicle from afar and human intervention is largely restricted to supervision and repair. Whilst this may happen before 2040, it is unlikely it will fully occur within the next few years, mostly due to the technical challenges and regulations.

Challenges faced by agricultural robotics

Compared to industries such as automotive, adoption of robotics and other digital technologies has been slow in agriculture, mainly due to agricultural robotics being immature. Farmers would not risk their harvest and entire yearly revenue at the expense of adopting a piece of machinery that does not demonstrate benefits.

A decline in need for labour is another issue. So, it does not look like the transition to robotics will be rapid. Restrictions caused by the Covid-19 pandemic, coupled with lower profits, may also prevent farmers from investing in agricultural robotics.

And even if agricultural robots are used on a large scale, the operating costs, including subscription of software packages and RTK GPS systems, will be quite high. Only skilled people would be able to operate and repair different robot parts.

> The US and Europe currently dominate robotics development. France has the highest proportion of agricultural robots, with over 10,000 robots (majority of which are milking robots or animal husbandry robots) already operating in farms across the country. Some active players in the agricultural robotics industry include John Deere (heavy machinery), Ecorobotix (smart weeding robots), and Traptic (fruit picking).

To mitigate the high upfront costs, robotics companies can offer a robot-as-a-service (RaaS) model where the farmer pays a subscription fee to the company that leases out the robot to the farmer. It would also mean that the farmers do not need to go through extensive training to operate the robot.

Attractive as it may seem, the RaaS model works only where there are trained operators nearby. Farms are often located in rural areas where there is no reliable 3G or 4G coverage. Robots need to be continuously connected to the internet for sending data in real time. It is doubtful whether robotic agricultural companies will be willing to get engineers or technicians out on the site for operating and repairing the robots.

What next?

Answers to most of the challenges stated above lie in the problems. The agricultural sector has always been stubborn to adopt technology. But with rising food shortages and declining labour supply, there is no other option left. Once robotics is embraced by the agricultural sector on a wider scale, its global market will rise and continuously do so for the next ten years. EFY

The article is based on IDTechEx webinars hosted by Dr Michael Dent and Yulin Wang, technology analysts at IDTechEx. It has been transcribed and curated by Vinay Prabhakar Minj, an IoT and audio electronics enthusiast at EFY.